Leonardo Iturriaga Tisoc

Lima, Perú

liturriaga@pucp.edu.pe leonarpro68@gmail.com leonarpro68-cmd.github.io GitHub

Research Interests

My interests lie in learning-based control for legged robots and mobile manipulators: training locomotion policies with reinforcement learning in simulation, making them robust to the limits of low-cost hardware, and bringing them to physical robots that I design and build myself, from CAD and fabrication to embedded software and ROS 2 integration.

Education

Pontificia Universidad Católica del Perú *2020 – Present*
BS in Mechatronics Engineering

- **Relevant Coursework:** Dynamic Systems and Control, Electric Circuits, Machine Elements, Manufacturing Processes, Materials Engineering, Thermofluids, Probability and Statistics.

Diploma in Biomechanics and Biomedical Engineering *Jun 2024 – Oct 2025*
P4H Bionics, Mexico (international program)

- Designed a bionic arm as the final project, applying biomechanical design and medical robotics.

Diploma in Digital Fabrication *Mar 2023 – Jun 2023*
FABRICUM, Pontificia Universidad Católica del Perú

- Specialization in 3D printing and digital design for industrial applications.

Research Experience

Simulation Lead – Summit XL Swarm (Swarmbots) *Lima, Peru*
Collaborative robotics research project, Pontificia Universidad Católica del Perú *2026 – Present*

- Migrated the swarm simulation from ROS 2 Humble / Gazebo Fortress to ROS 2 Jazzy / Gazebo Harmonic (real-time factor ≈1.0) and set up six worlds to diversify the dataset.
- Simulated swarm of omnidirectional Summit XLS robots with grippers and RGB-D cameras that collects trash and sorts it by color into depots (3 robots, 6/6 pieces, 0 collisions).
- Built a rosbag2 → HDF5 pipeline in ACT/ALOHA format to train imitation-learning policies from teleoperated demonstrations.
- Tools Used: ROS 2, Gazebo Harmonic, Docker, Python. [code]

Projects

Bipedo: 12-DOF Bipedal Robot Trained with Reinforcement Learning [video] [code]

- Designed and 3D-printed a 1.57 kg, 0.46 m biped with 12 Feetech STS3215 servos, a Raspberry Pi 5 and an IMU; a symmetric torso redesign reduced torso mass by 18% and pitch inertia about the hip axis by 69%.
- Wrote a CAD-to-MJCF generator that builds the MuJoCo model part by part from the Inventor assembly, with measured masses, joint ranges and servo axes; identified the servo model from 97 public test runs.
- Trained a velocity-tracking walking policy in mjlab (4096 parallel environments, RTX 4060): 87–91% forward speed tracking, 99.6% on turns, no falls. Ran the same policy sim-to-sim in Gazebo Harmonic through a C++ gz-sim plugin.
- Built and tested the real right leg with servo telemetry (load, current, temperature); showed that walking torque peaks (2.3–2.4 N m) exceed the 1.91 N m of the 7.4 V servo.
- Tools Used: Inventor, MuJoCo/mjlab, Gazebo, Python, C++, 3D Printing.

Pentabot: Five-Legged Robot Trained to Walk Without Encoders (team project) [video] [code]

- Modeled the 15-DOF robot in MuJoCo from the Inventor CAD and trained a PPO locomotion policy that uses only the IMU, the command and its own past actions, so it can run on MG995 servos without position feedback.
- Across 256 domain-randomized robots (servo gain, torque, delay, backlash, IMU noise): 83–89% speed tracking and 0% falls, versus 54–86% and 6–10% for an encoder-based baseline.
- Kept the network to ≈84k parameters for deployment on an ESP32 at 50 Hz.
- Tools Used: MuJoCo/mjlab, PPO, Python, ESP32.

ArmCar: Mobile Manipulator with a Voice-Controlled AI Agent [video] [code]

- Designed a distributed ROS 2 system (PC as brain, Raspberry Pi 5 as body) with 8 packages: DualSense teleoperation in joint space and with inverse kinematics, differential drive with a safety watchdog, servo-bus driver, OLED face and camera.
- Integrated an LLM agent (Claude) with local speech (wake word, Whisper, Piper) that moves the real SO-101 arm through joint limits and interpolation, and describes the scene from a webcam.
- Tools Used: ROS 2 Jazzy, Python, RViz, Raspberry Pi 5, Claude API.

Teaching and Professional Experience

Co-founder and Head Instructor *Cusco, Peru*
Jaque Matheus (robotics academy) *Jan 2025 – Feb 2025*

- Co-founded a robotics academy for children aged 6–12; independently designed and taught the introductory robotics curriculum.

Physics and Chemistry Teacher *Cusco, Peru*
Colegio Fleming *Jan 2025 – Feb 2025*

- Taught secondary-level physics and chemistry.

Training Courses – Pontificia Universidad Católica del Perú

Robotic Manipulators *Jan 2026 – Feb 2026*
Autonomous Navigation of Mobile Robots *Nov 2025 – Jan 2026*
Robot Operating System (ROS 2) *Oct 2025 – Nov 2025*
Linux and Python Fundamentals *Sep 2025 – Oct 2025*

Technical Skills

Programming: Python, MATLAB, Simulink, C/C++ (basic), Linux, Bash, Git, Docker
Robotics Frameworks: ROS 2 (Humble, Jazzy), MuJoCo, mjlab, Gazebo (Fortress, Harmonic), RViz, URDF / MJCF / SDF
Robot Learning: Reinforcement learning (PPO), domain randomization, sim-to-sim transfer, ONNX, imitation-learning data (ACT)
AI Integration: LLM agents with tool use (Claude API), Whisper, Piper
Software: Autodesk Inventor, Fusion 360, 3D printing (PLA, TPU, multi-material)
Embedded Platforms: Raspberry Pi 5, Raspberry Pi Pico, NVIDIA Jetson Nano, ESP32, Arduino
Languages: Spanish: Native, English: B2 (Upper-Intermediate)